

Exercise 02

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1 Rounding for Bin Packing

For the construction of a PTAS for the Bin packing problem we needed a tricky way of rounding. Here we investigate why this is necessary.

Construct a set of items with sizes $0 < x_1, \dots, x_n < 1$ with the following property: while it is possible to pack these into some number m of bins, if we for any $\varepsilon > 0$ consider the set of items with sizes $(1 + \varepsilon)x_1, \dots, (1 + \varepsilon)x_n$, we need at least αm bins for some $\alpha > 1$. What is the best α you can get and how does it relate to the issue of designing a PTAS for Bin packing?

2 Target shooting

Suppose we have a ground set S and we wish to estimate the size of a subset $T \subseteq S$. We can do it by repeatedly sampling from S : we sample m times a uniform element from S and let X_i to be an indicator for the event of the i -th sample being an element of T . One can use the Chernoff bound to prove that the variable $(X_1 + X_2 + \dots + X_m)/m$ is within $(1 + \varepsilon)$ multiplicative error of $|T|/|S|$ with probability at least $1 - \delta$ if we set

$$m = \Theta\left(\frac{|S|}{|T|} \varepsilon^{-2} \log(1/\delta)\right).$$

1. Argue that the lemma really follows from the Chernoff bound.
2. Argue that if we sample only $O(\frac{|S|}{|T|})$ elements, with constant probability, we never sample an element from T throughout the whole procedure and, hence, we cannot get a reasonable estimate.
3. Suppose that we have a FPRAS that returns value that is in the correct range $[(1 - \varepsilon)OPT, (1 + \varepsilon)OPT]$ with probability $2/3$. Recall the median trick from the lecture and show how to amplify the probability to $1 - \delta$ with $O(\log 1/\delta)$ calls of the original algorithm.

3 Counting satisfying assignments

In the lecture we have seen a FPRAS that estimates the number of satisfying assignments of a given DNF formula. Suppose that we wish to solve this problem and we know that one of the clauses contains only 10 variables. Can you propose a faster algorithm than the one you know from the lecture?

4 Cut Counting

Suppose we are given a multigraph $G = (V, E)$ and an edge e with endpoints u and v . We define G/e as the contraction of G along e by removing the vertices u and v and replacing them with a single super-vertex, s . Any edge in G that was incident to u or v will instead be incident to s in G/e . We discard any self-loops that form during this process, but retain any parallel edges that are created.

Karger's algorithm is a simple randomized algorithm for finding the minimum cut of a graph. It proceeds as follows: starting with the graph G , we iteratively choose an edge uniformly at random and contract along it. This process stops when exactly two vertices remain, and the algorithm returns the size of the cut separating them.

We have claimed in the lecture that in each graph with minimum cut size k , for any $\alpha \geq 1$, the number of cuts of size at most αk is at most $n^{2\alpha}$. We want to show this via Karger's algorithm. Fix any cut C of size at most αk . Apply Karger's algorithm, but stop when the number of vertices reaches $r := \lceil 2\alpha \rceil$. Then, partition the remaining vertices uniformly at random into two non-empty sets A and B . Show that with probability at least $q := \frac{1}{2^{r-1}} \prod_{i=r+1}^n (1 - \frac{2\alpha}{i})$, the resulting cut equals C .

Conclude that there are at most $1/q$ cuts of size at most αk . Finally, show that if α is a positive integer, then $q \geq n^{-2\alpha}$. This last inequality is also true for general $\alpha \geq 1$, but harder to show, so we only ask you for the integer case.